

Coursework for Algorithms for Data Analysis and Data Mining (I+II)

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A. Data Analysis and Regression

1 Logistic Regression

Data provided for analysis with logistic regression was placed in a table below.

Table 1: Numbers of hours on programming and final result.

Hours	25	35	40	50	55	60	70	89	92	100
Result	0	0	0	1	0	1	1	1	1	1

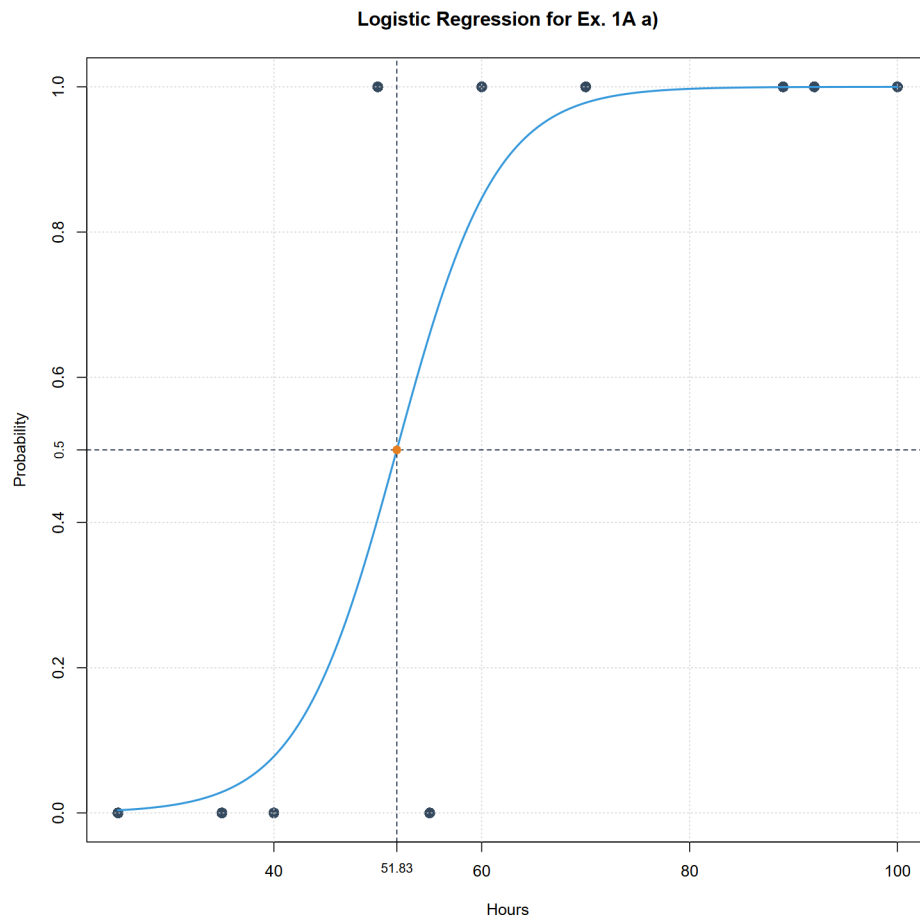
a) For this exercise we used code from file "Zad1Aa.R".

Summarizing the logistic regression for provided data set we receive following parameters:

Variable	Estimate	Std. Error	z value	<i>p</i> -value
Intercept	-10.8360	8.3232	-1.302	0.193
x	0.2091	0.1576	1.327	0.185

As we may see, basing on data from table 2, the *z* values for both variables are rather high what is proven by *p*-value being higher than 0.1 what can be considered as quite large. It is indeed result of low amount of data - 10 records equivalent to 9 degrees of freedom. This means that significance of our model is low thus statistically it is not important. Nevertheless, we may conclude that direction of dependence between hours of programming and final result is positive.

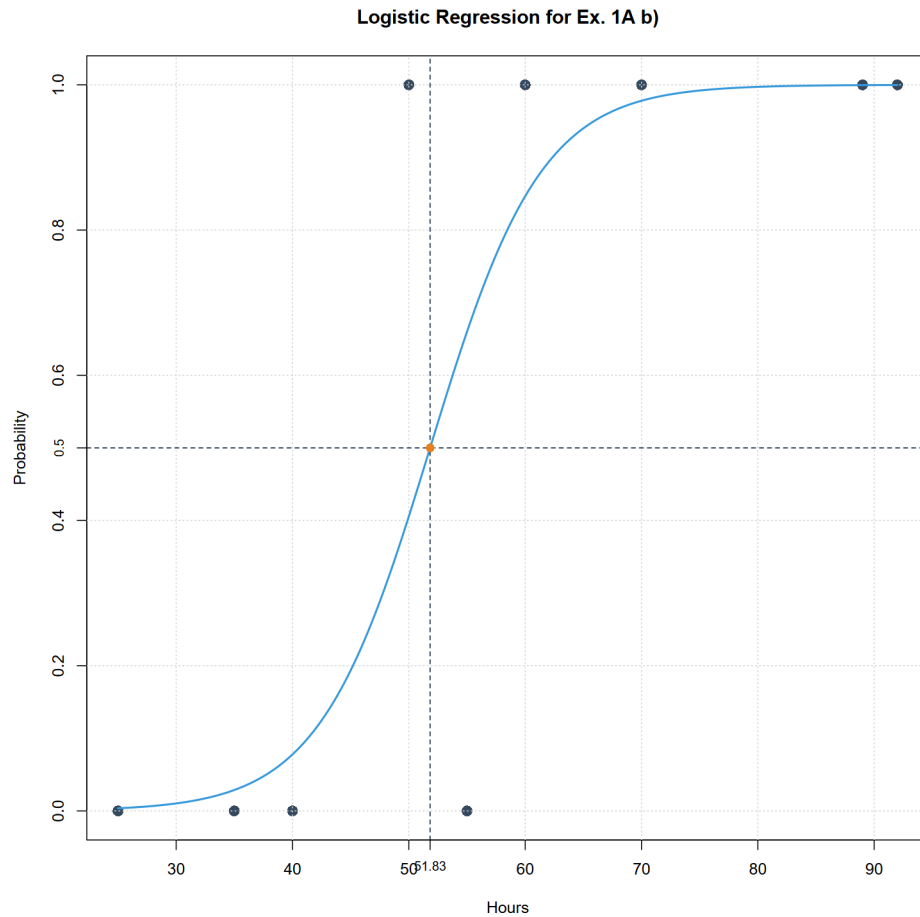
We may also plot the graph below.



- b) For this exercise we used code from file "Zad1Ab.R".
 Summarizing the logistic regression for provided data set, excluding point (100, 1),
 we receive following parameters:

Variable	Estimate	Std. Error	z value	<i>p</i> -value
Intercept	-10.8360	8.3232	-1.302	0.193
x	0.2091	0.1576	1.327	0.185

As we may see value of all parameters is the same as from the a) sub-point.

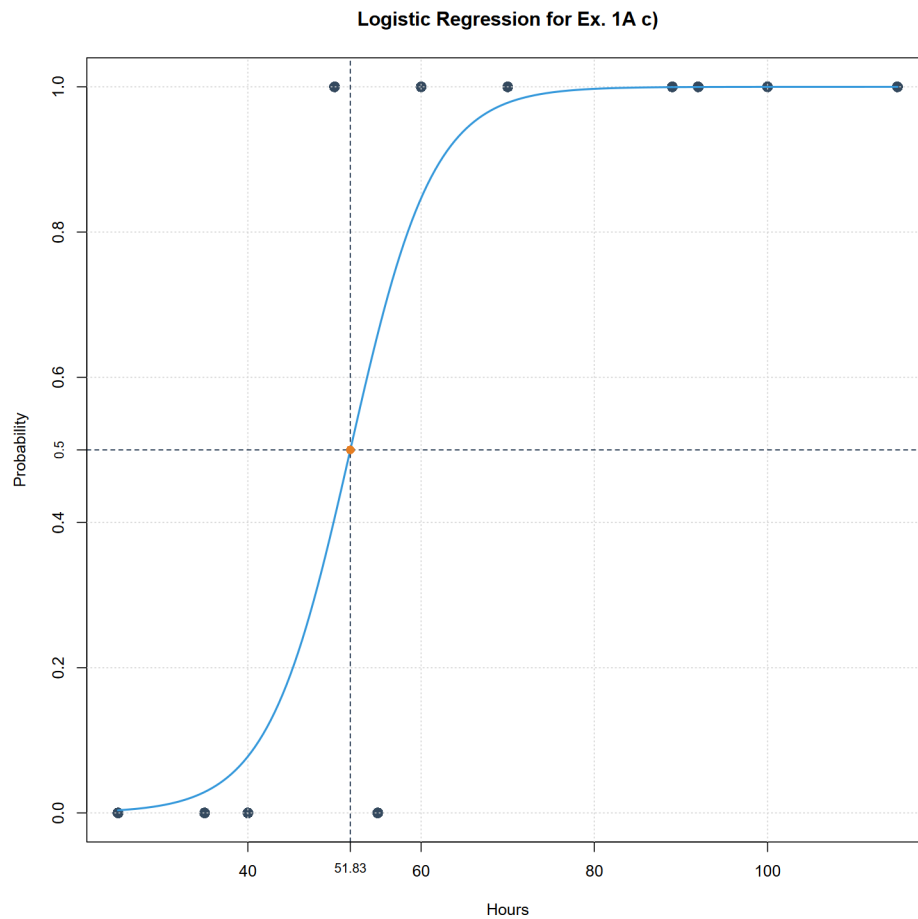


Moreover, the graph is also the same. Thus, we may conclude that removing point (100, 1) does not influence this model. We may suppose that it influenced by almost perfect correspondence of coordinates of this point with proposed model.

- c) For this exercise we used code from file "Zad1Ac.R".
Summarizing the logistic regression for data set from sub-point a, including new point (115, 1), we receive following parameters:

Variable	Estimate	Std. Error	z value	<i>p</i> -value
Intercept	-10.8360	8.3232	-1.302	0.193
x	0.2091	0.1576	1.327	0.185

As we may see value of all parameters is the same as from the a) and b) sub-points.



Moreover, the graph is also the same. This is not surprising as it confirms hypothesis from sub-point b) that when coordinates is almost perfectly corresponding with our model. This we may conclude, that the model would only change if value on y axis were 0 which we would then consider as outlier. But ones placed on x axis to the right and keeping $y=1$ does not affect model.

d) To sum up all above we may propose advantages and disadvantages

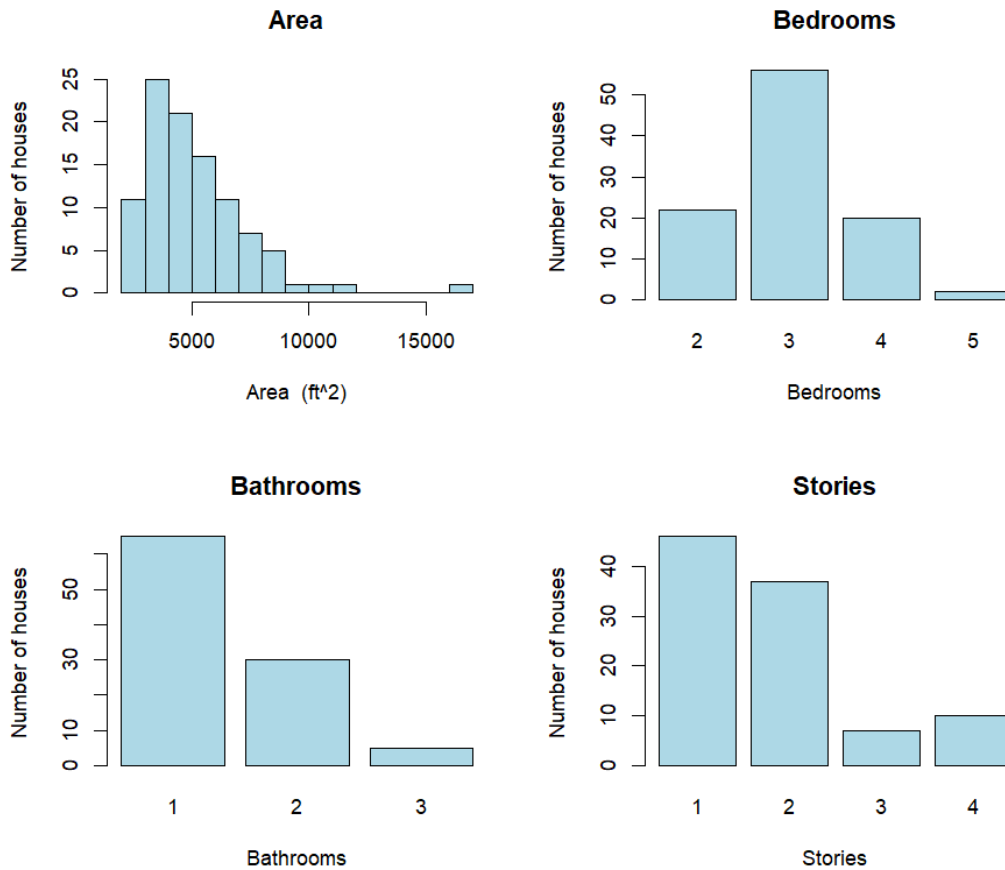
Advantages	Disadvantages
Simple algorithm	Very sensitive to outliers

B. kNN Regression Analysis

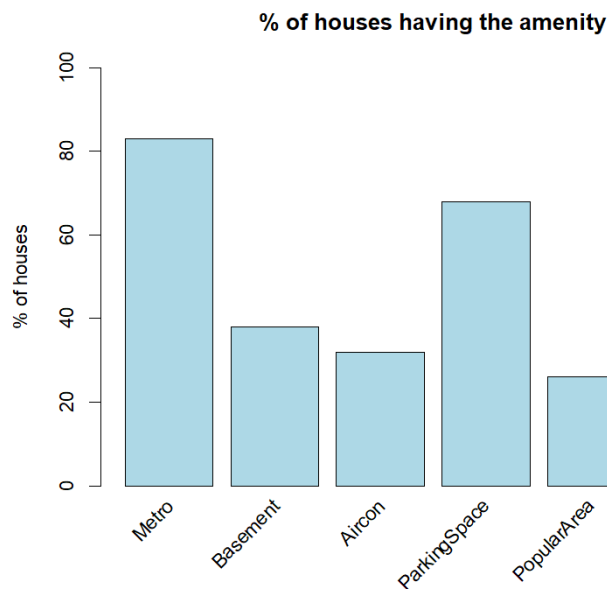
1 Summary of the sggw.b data

Here we used code from Zad2.R file. We will provide statistical analysis for the data. We have 10 explanatory features and one target feature - the price of the house. In the explanatory features (X-features) we have the Area, the numbers of Bathrooms, Bedrooms etc, and binary features telling us if there is metro in close proximity, is the place furnished, or is it a popular area.

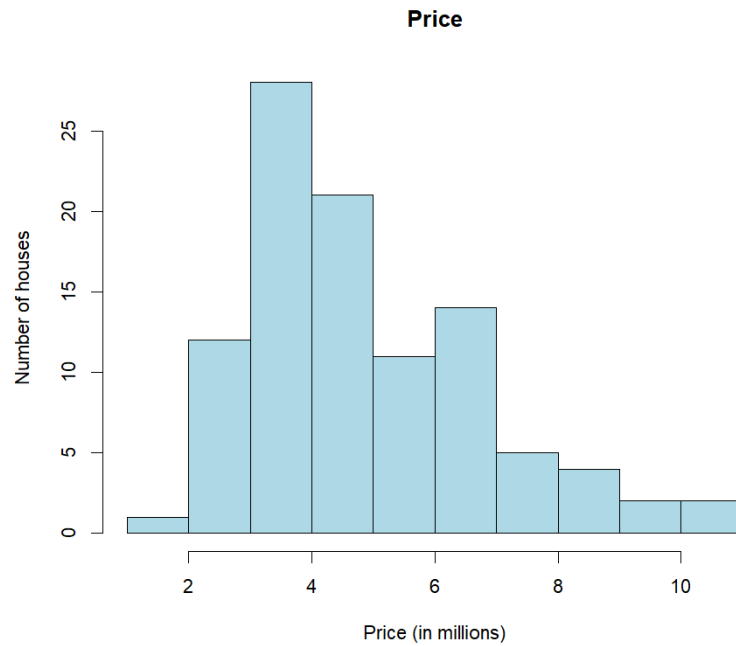
	Mean	SD	Min	Max
Area..ft2.	5167.32	2186.64	2145.00	16 200.00
Bedrooms	3.02	0.71	2.00	5.00
Bathrooms	1.40	0.59	1.00	3.00
Stories	1.81	0.95	1.00	4.00
Metro	0.83	0.38	0.00	1.00
Basement	0.38	0.49	0.00	1.00
Aircon	0.32	0.47	0.00	1.00
ParkingSpace	0.68	0.87	0.00	3.00
PopularArea	0.26	0.44	0.00	1.00
Furnished	0.42	0.50	0.00	1.00
Price (in mln.)	4.88	1.93	1.75	10.85



We can see that the area is similar to the Chi-squared distribution, the peak is around 3000 square feet. Most houses have 3 bedrooms, 1 bathroom and 1 or 2 stories



Most of the houses has a metro station nearby at over 80 percents, and a parking space at 70. The place being furnished or having a basement has around 40 percent chance. The least chance is to get an air conditioning or being in the popular area.



The price has a distribution similar to the area distribution.

2 kNN regression model

Why did we choose kNN? It is perfect for predicting the value of the houses, as it chooses the most similar places, and checks the price with them. We used FNN (Fast Nearest Neighbor Search Algorithms and Applications) library, in which it is possible to do regression on k nearest neighbours model